

Multiple Choice (10 marks)

1. __ refers to a model that can neither model the training data nor generalize to new data.
 - (a) good fitting
 - (b) overfitting
 - (c) underfitting
 - (d) all of the above
2. Which of the following is NOT supervised learning?
 - (a) PCA
 - (b) Decision Tree
 - (c) Linear Regression
 - (d) Naive Bayesian
3. Which image data augmentation is most common for natural images?
 - (a) random crop and horizontal flip
 - (b) random crop and vertical flip
 - (c) posterization
 - (d) dithering
4. Predicting the amount of rainfall in a region based on various cues is a _____ problem.
 - (a) Supervised learning
 - (b) Unsupervised learning
 - (c) Clustering
 - (d) None of the above
5. Which of the following tasks can be best solved using Clustering.
 - (a) Predicting the amount of rainfall based on various cues
 - (b) Detecting fraudulent credit card transactions
 - (c) Training a robot to solve a maze
 - (d) All of the above

True / False (10 marks)

Answer True or False

6. True or False: The original ResNet paper utilized Batch Normalization rather than Layer Normalization.
7. True or False: The original ResNet incorporates self-attention mechanisms, similar to the original Transformers.
8. True or False: Applying an L_1 norm regularization penalty in linear regression can result in some coefficients being zeroed out.
9. True or False: The joint probability $P(H, U, P, W)$ can be expressed as $P(H) * P(W) * P(P | W) * P(U | H, P)$.
10. True or False: The classification run time of the nearest neighbors algorithm is $O(\log N)$, which implies efficient searching within the dataset.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the choice of polynomial degree in polynomial regression impacts the balance between underfitting and overfitting, and analyze the implications for model performance.
12. Discuss the importance of centering data to achieve equivalent projections in Principal Component Analysis (PCA) and Singular Value Decomposition (SVD). Analyze the implications for data interpretation.

End of Paper